Princess Sumaya University for Technology
Communication Engineering Department
Electromagnetics I
Worksheet 4

1. A circular ring of radius a carries a uniform charge ρ_L C/m and is placed on the xy-plane with axis the same as the z-axis. Show that:

$$\boldsymbol{E}\,(0,0,\mathrm{h}) = \frac{\rho_L a h}{2\varepsilon_o[h^2 + a^2]^{3/2}}\,\boldsymbol{a_z}$$

a. What values of h gives the maximum value of $\boldsymbol{E}$**?**
b. If the total charge on the ring is Q, find $\boldsymbol{E}$ as $a \to 0$.

2. For a spherical charge distribution:

$$\rho_v = \begin{cases} \rho_o(a^2 - r^2), & r < a \\ 0, & r > a \end{cases}$$

a. Find $\boldsymbol{E}$ and V for $r \geq a$**.**
b. Find $\boldsymbol{E}$ and V for $r \leq a$**.**
c. Find the total charge.
d. Show that $\boldsymbol{E}$ is maximum when $r = 0.145\,a$**.**